2-5 TIME SERIES PLOTS

- **Limitations of standard plots:** Tools like histograms, stem-and-leaf plots, and box plots show variability but ignore the influence of time.

- **Definition:** A **time series** is a data set where observations are recorded specifically in the order they occur.

- **Visual representation:** A **time series plot** graphs the observed values on the vertical axis against time (minutes, days, years, etc.) on the horizontal axis.

- **Purpose:** Plotting data this way reveals trends, cycles, and patterns that are otherwise invisible in non-temporal displays.

- How different views of data reveal specific patterns over time:

  - **Time Series**: A **time series** is a data set where observations are recorded in the order they occur.

  - **Time Series Plot**: A **time series plot** is a graph that plots the observed value (vertical axis) against time (horizontal axis) to reveal trends or cycles.

  - **Annual View**: In **Fig. 2-15**a, a 10-year plot shows a general upward **trend** despite annual variability.

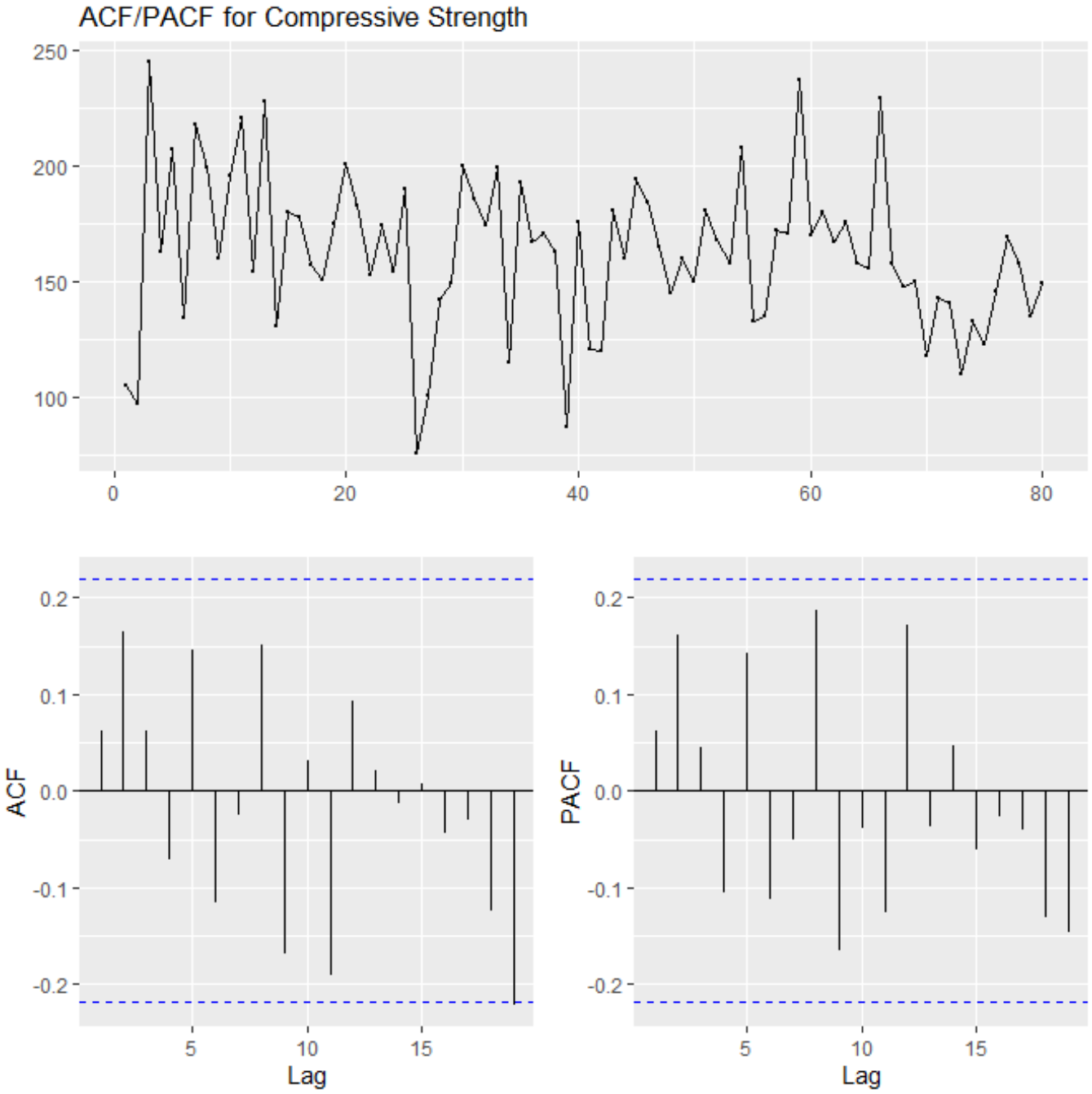
- **Quarterly View**: In **Fig. 2-15**b, a 3-year plot by quarter reveals **cyclic** variability, with higher sales in the first and second quarters.

- Combining a **time series plot** with other graphical displays can provide more insight into data variability:

  - **Integrated Method**: J. Stuart Hunter, writing in **The American Statistician**, proposed merging a stem-and-leaf plot with a **time series plot**.

  - **Digidot Plot**: This specific combination is known as a **digidot plot**, which allows for the simultaneous observation of data trends over time and the overall distribution.

- A visual analysis of compressive strength data:
  - **Figure 2-16** displays a digidot plot based on the data from Table 2-2, assuming the observations are in chronological order.
  - The plot serves a dual purpose: it shows both the overall variability of the data and how that variability changes over time.
  - The data typically fluctuates around a mean value of 162.67.
  - There is no strong or obvious **trend** or **cyclic** pattern present in the variability over time.


ACF/PACF for Compressive Strength
250
200
150
100
0
20
40
60
80
ACF
0.2
0.1
0.0
-0.1
-0.2
5
10
15
Lag
PACF
0.2
0.1
0.0
-0.1
-0.2
5
10
15
Lag

- **Fig. 2-17** illustrates chemical process concentration readings over time.

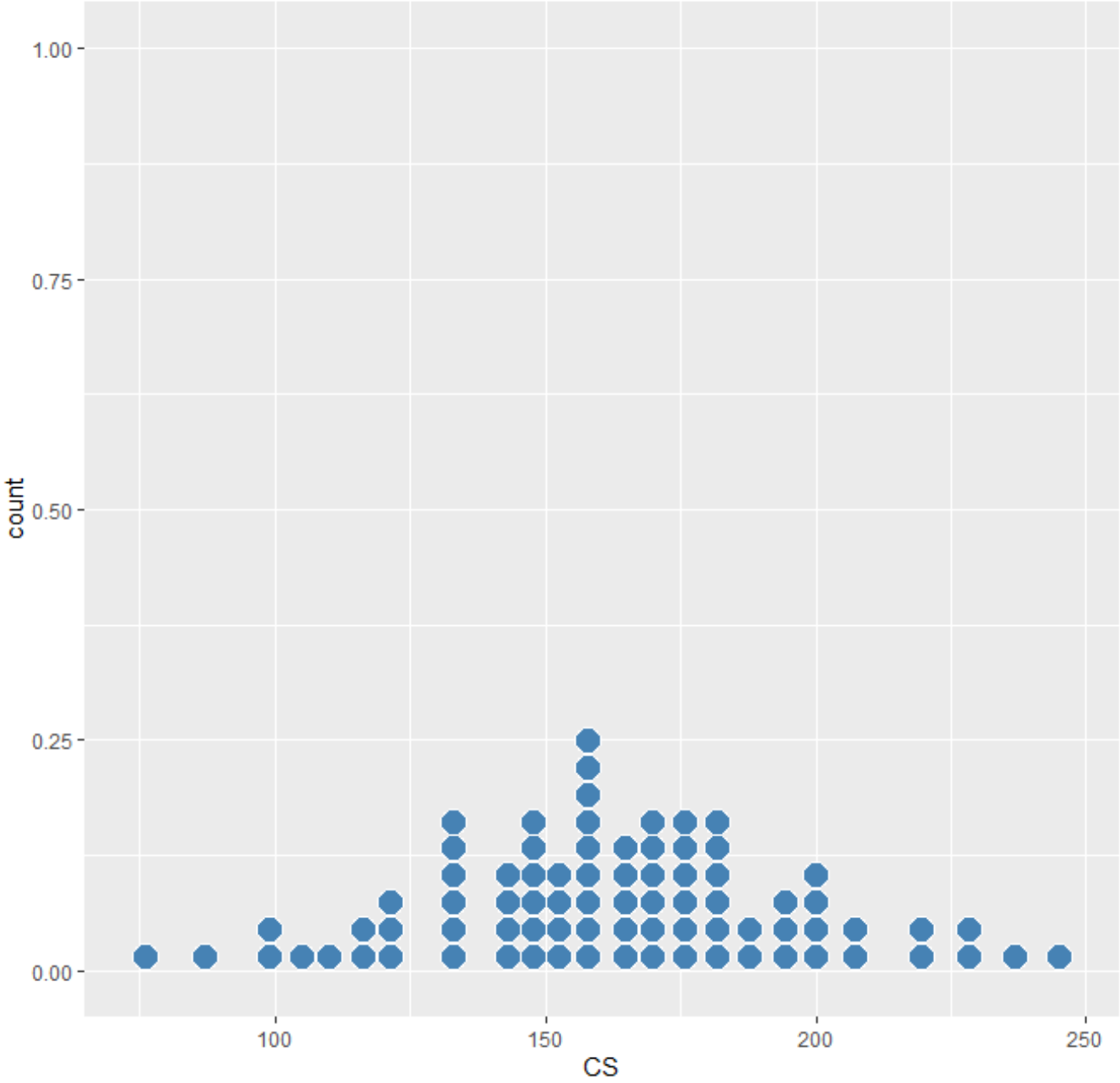
  - **Data Source:** Displays 30 hourly observations of chemical product concentration.
  - **Initial Performance:** During the first 20 hours, concentrations were consistently above 85 g/l.
  - **Process Shift:** After the 20th sample, a potential issue occurred, leading to a visible drop in concentration levels.

- **Improvement Goal:** Reducing this output variability is identified as the primary way to improve process operation.

- **Context:** The data in **Fig. 2-17** is also represented as a control chart in Fig. 1-19 of Chapter 1.

- **Figure 2-18** displays a Minitab **marginal plot** for the compressive strength data, offering a similar, commonly supported alternative to the digidot plot. This visualization displays data in observation order, featuring a histogram, box plot, or dot diagram in the margins of the main graph.

Figure 2-18: Marginal Plot (Dotplot of Compressive Strength)



2-48. The annual Wolfer sunspot numbers from 1770 to 1869 are shown in Table 2-7. (For an interesting analysis and interpretation of these numbers, see the book by Box and Jenkins referenced in Exercise 2-47. The analysis requires some advanced knowledge of statistics and statistical model building.)

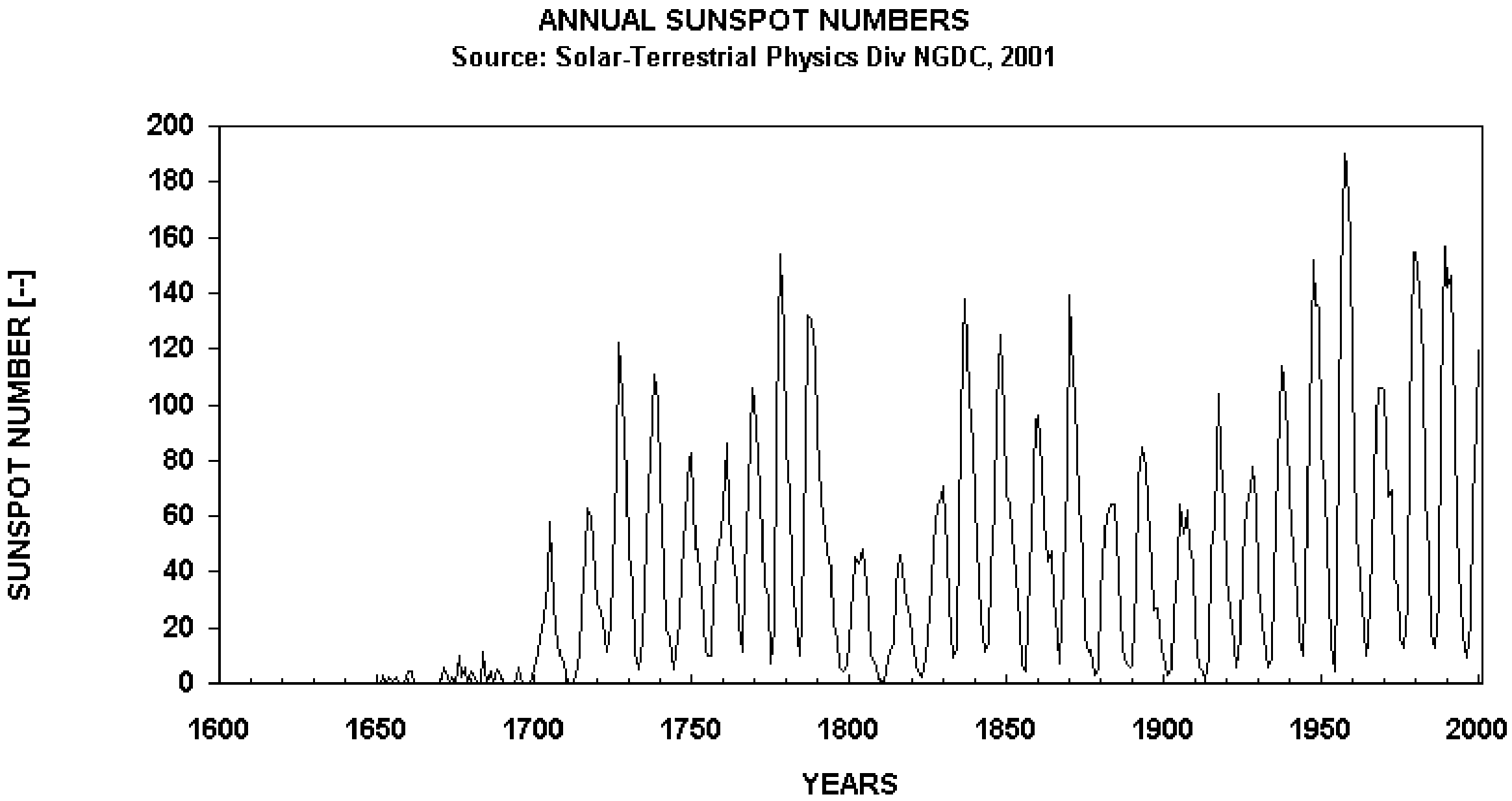
(a) Construct a time series plot of these data.

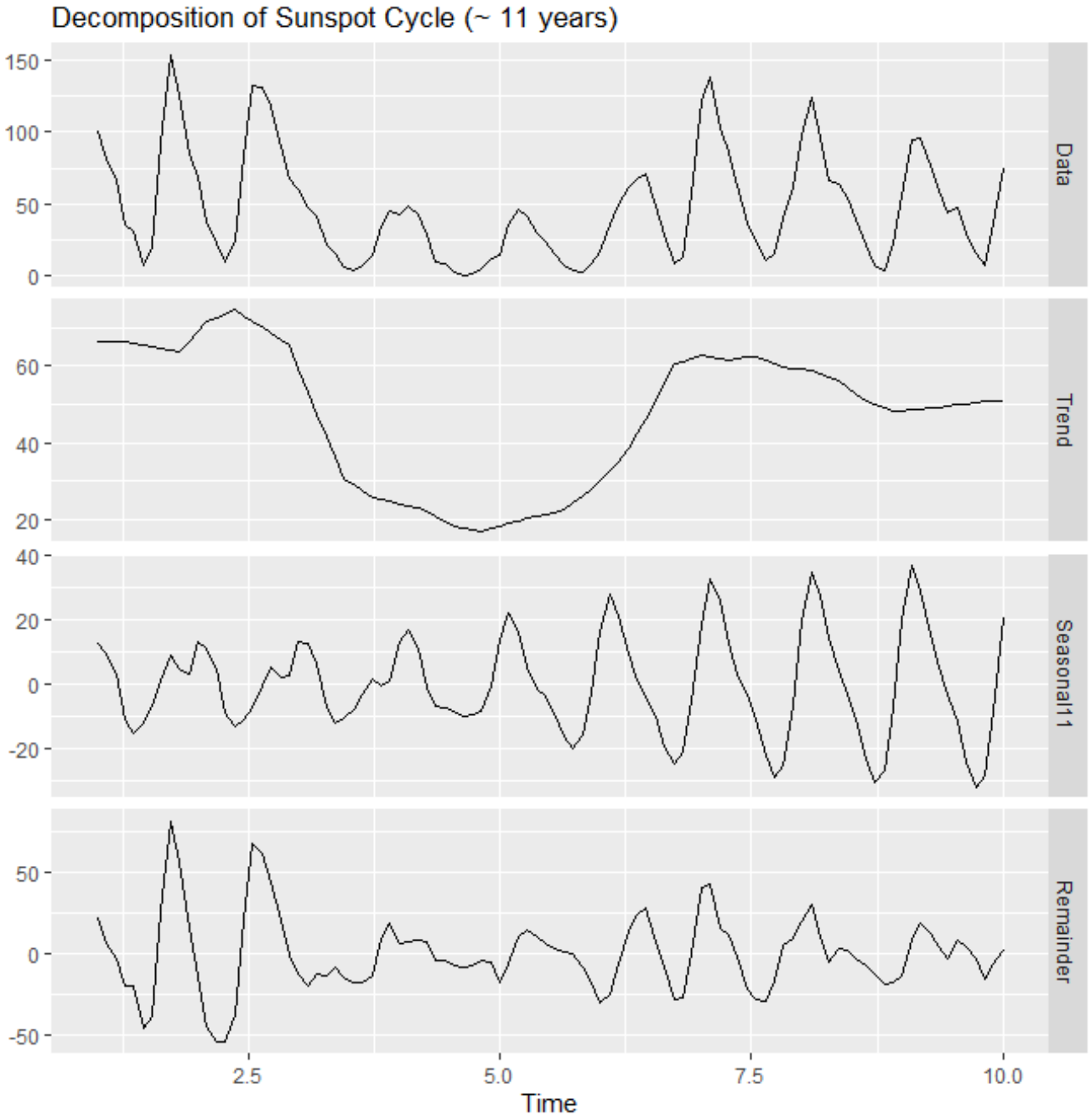
(b) Construct and interpret either a digidot plot (or a marginal plot) or a stem-and-leaf and time series plot of these data.

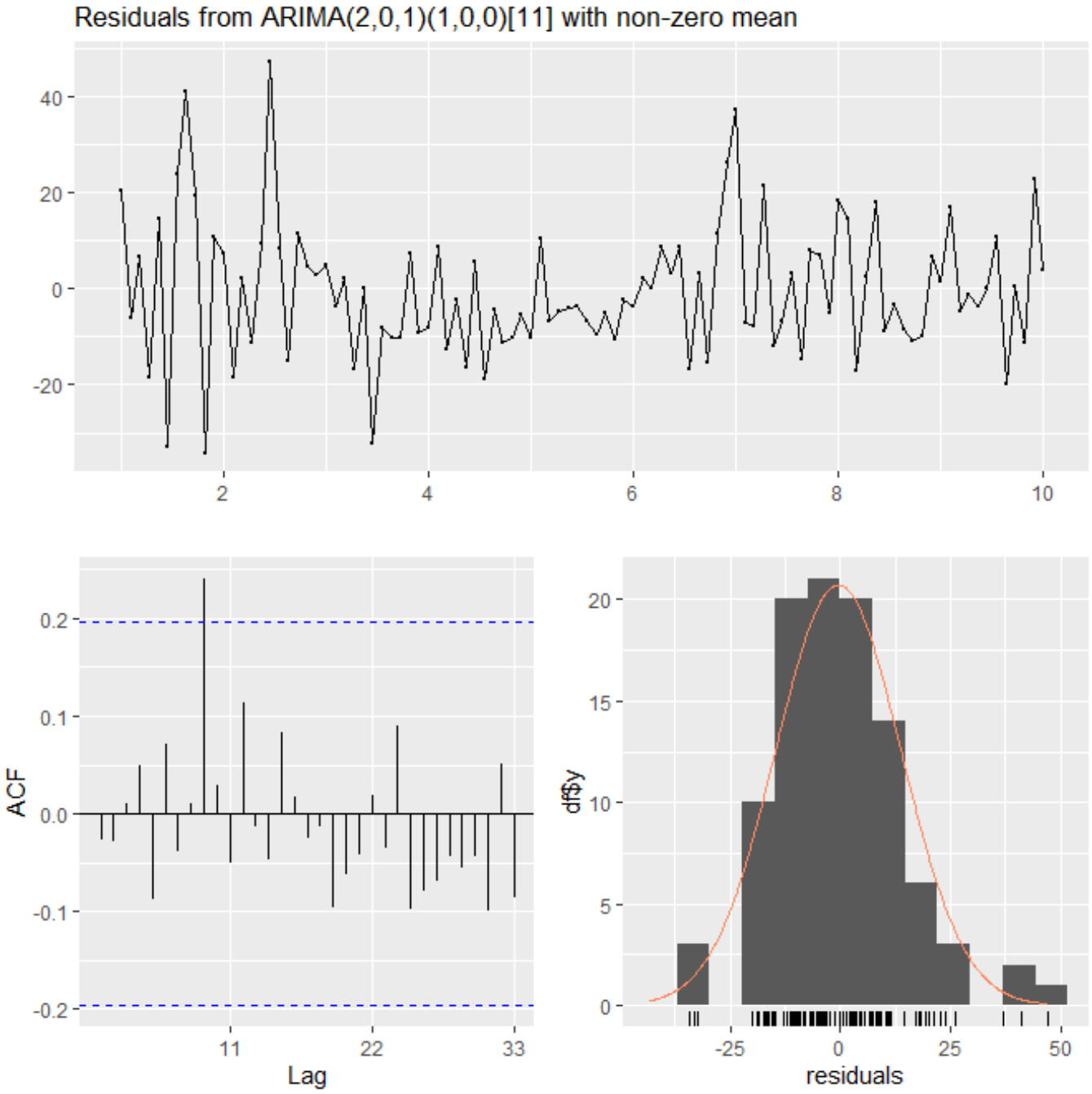
2-47. In their book *Time Series Analysis, Forecasting, and Control* (Holden-Day, 1976), G. E. P. Box and G. M. Jenkins present chemical process concentration readings made every 2 hours. Some of these data are shown next. (Read down, then left to right.)

17.0	16.6	16.3	16.1	17.1	16.9	16.8	17.4
17.1	17.0	16.7	17.4	17.2	17.4	17.4	17.0
17.3	17.2	17.4	16.8	17.1	17.4	17.4	17.5
17.4	17.6	17.4	17.3	17.0	17.8	17.5	18.1
17.5	17.4	17.4	17.1	17.6	17.7	17.4	17.8
17.6	17.5	16.5	17.8	17.3	17.3	17.1	17.4
16.9	17.3						

Construct and interpret either a digidot plot or a stem-and-leaf and time series plot of these data.


ANNUAL SUNSPOT NUMBERS
Source: Solar-Terrestrial Physics Div NGDC, 2001
SUNSPOT NUMBER [--]
200
180
160
140
120
100
80
60
40
20
0
1600
1650
1700
1750
1800
1850
1900
1950
2000
YEARS

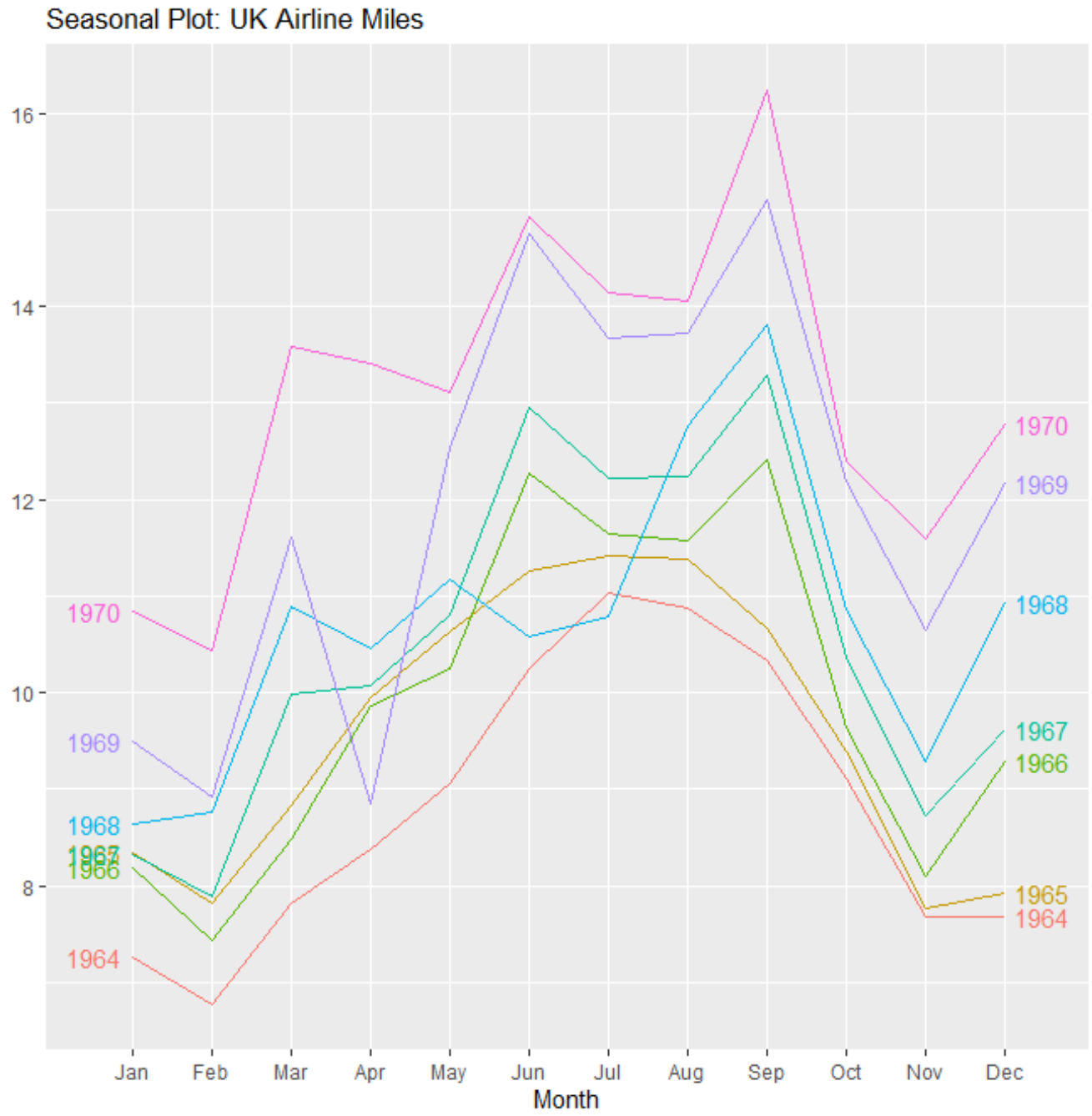

Decomposition of Sunspot Cycle (~ 11 years)
150
100
50
0
Data
60
40
20
Trend
40
20
0
-20
Seasonal11
50
0
-50
Remainder
2.5
5.0
7.5
10.0
Time

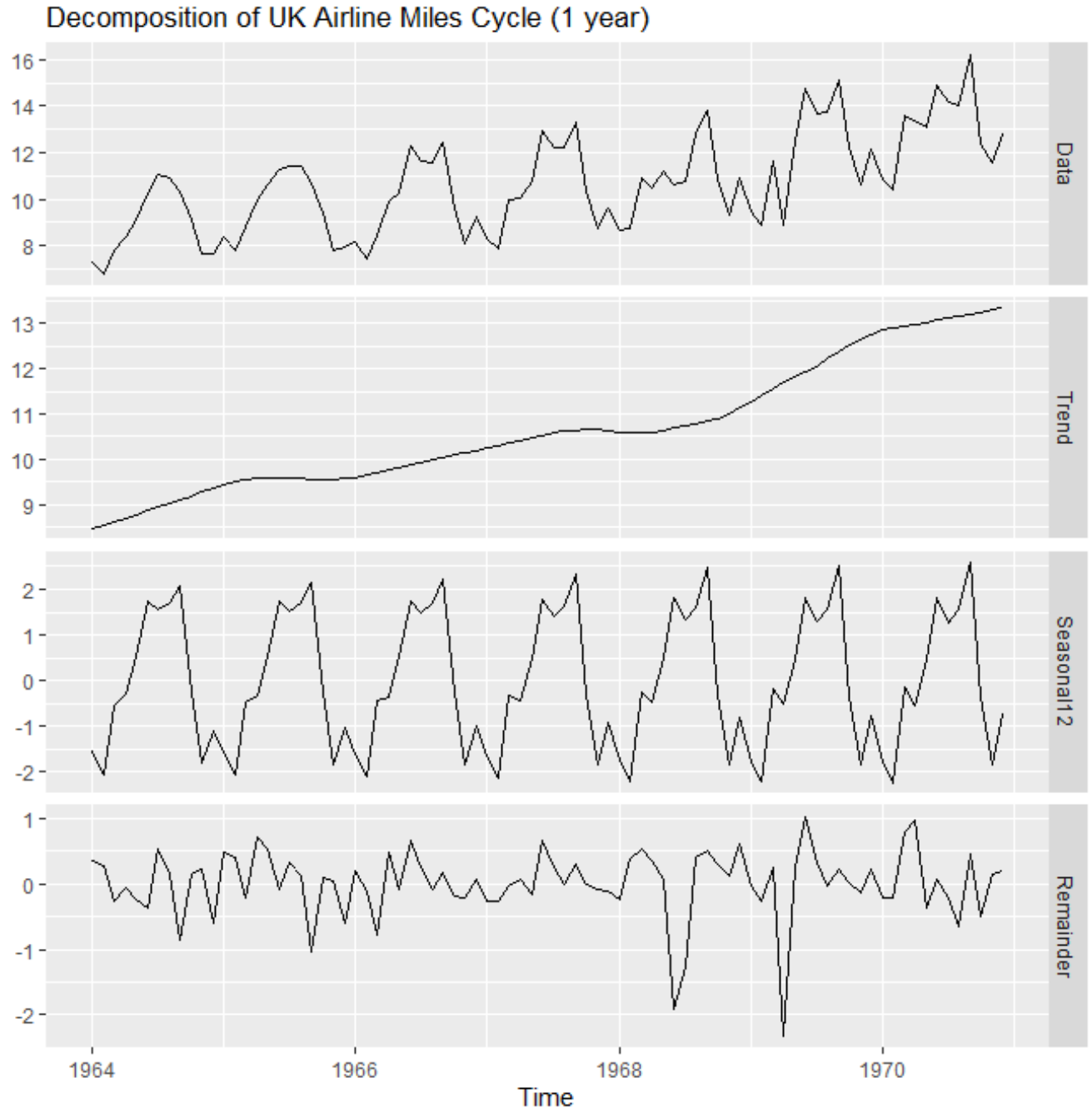

Residuals from ARIMA(2,0,1)(1,0,0)[11] with non-zero mean
40
20
0
-20
2
4
6
8
10
0.2
0.1
0.0
-0.1
-0.2
ACF
11
22
33
Lag
20
15
10
5
0
df$y
-25
0
25
50
residuals

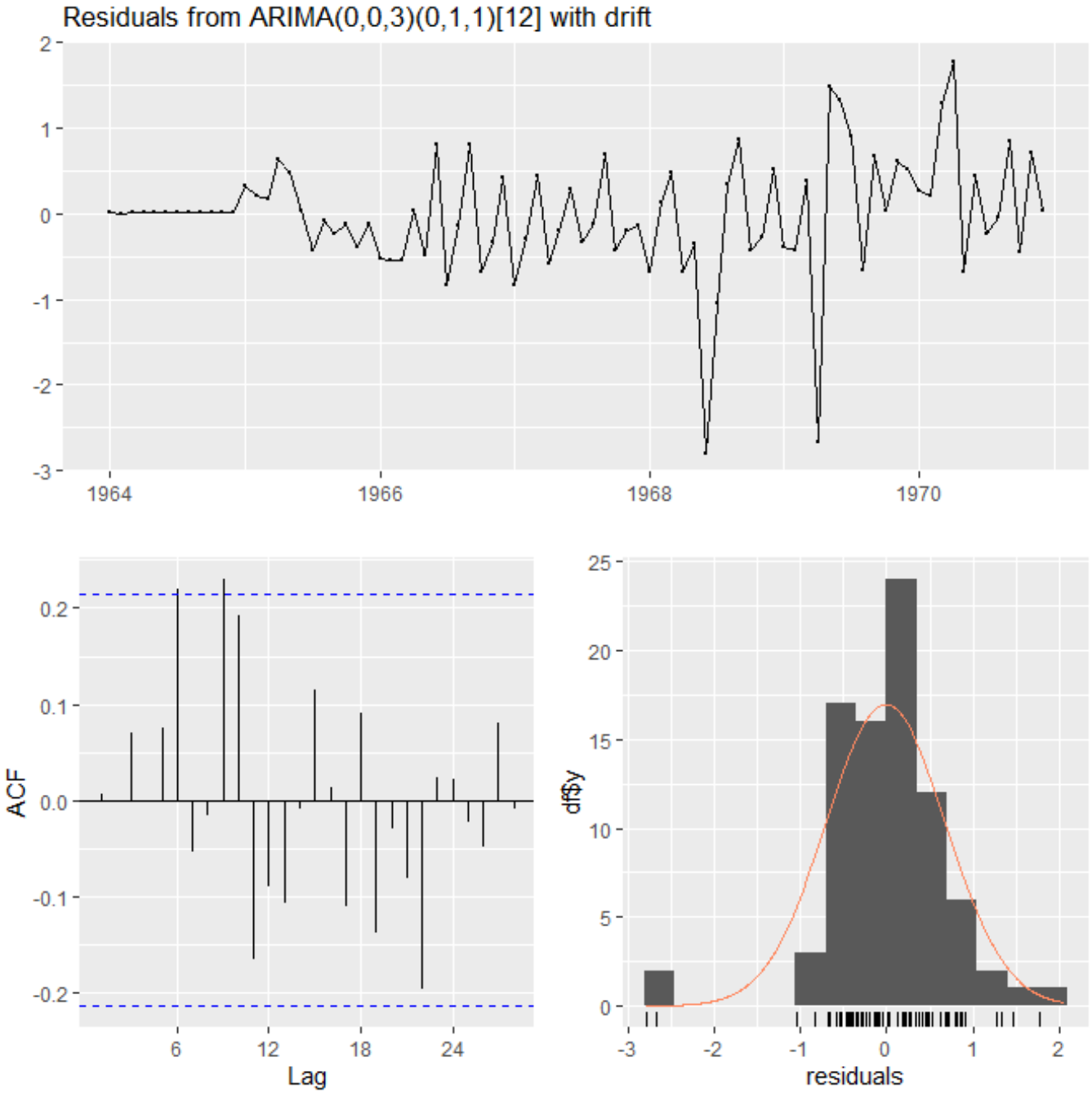
2-49. In their book *Forecasting and Time Series Analysis*, 2nd ed. (McGraw-Hill, 1990), D. C. Montgomery, L. A. Johnson, and J. S. Gardiner analyze the data in Table 2-8, which are the monthly total passenger airline miles flown in the United Kingdom, 1964–1970 (in millions of miles).

(a) Draw a time series plot of the data and comment on any features of the data that are apparent.

(b) Construct and interpret either a digidot plot (or a marginal plot) or a stem-and-leaf and time series plot of these data.


Seasonal Plot: UK Airline Miles
16
14
12
10
8
1970
1969
1968
1967
1966
1964
1965
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month


Decomposition of UK Airline Miles Cycle (1 year)
Data
Trend
Seasonal12
Remainder
16
14
12
10
8
13
12
11
10
9
2
1
0
-1
-2
1
0
-1
-2
1964
1966
1968
1970
Time


Residuals from ARIMA(0,0,3)(0,1,1)[12] with drift
2
1
0
-1
-2
-3
1964
1966
1968
1970
ACF
0.2
0.1
0.0
-0.1
-0.2
6
12
18
24
Lag
df$y
25
20
15
10
5
0
-3
-2
-1
0
1
2
residuals